

Topology First-Year Exam, January 2024, Part 1

For the first five problems, show all your work and support all statements.

1. Prove that $\{0, 1\}^\omega$ is not countable.
2. Show that $\mathbb{R}^\omega$ with the box topology is not connected.
3. Assume that $f : X \rightarrow Y$ is continuous and surjective.
 - (a) If X is Lindelöf (every open covering has a countable subcovering), show that Y is also.
 - (b) If X is separable (has a countable dense subset), show that Y is also.
4. Let (X, d) be a compact metric space. Let $f : X \rightarrow X$ and suppose there is some $c < 1$ with $d(f(x), f(y)) \leq c \cdot d(x, y)$ for all $x, y \in X$. Prove there exists a unique $x \in X$ with $f(x) = x$.
5. Let (Y, d) be a complete metric space. Let J be a set. Suppose $d(y, y') \leq 1$ for all $y, y' \in Y$, meaning a simpler definition for the uniform metric $\bar{\rho}$ on Y^J is

$$\bar{\rho}(f, g) = \sup_{\alpha \in J} \{d(f(\alpha), g(\alpha))\}$$

for $f, g : J \rightarrow Y$. Prove that $(Y^J, \bar{\rho})$ is a complete metric space.

Answer the following problems with complete definitions, complete statements, an example, or a short proof.

6. Define what it means for a topological space X to be regular.
7. Show that a closed subspace C of a compact space X is compact.
8. A function $f : \mathbb{Z}_+ \rightarrow \{0, 1\}$ is eventually zero if there is a positive integer N such that $f(n) = 0$ for all $n \geq N$. Prove that the set of all functions $f : \mathbb{Z}_+ \rightarrow \{0, 1\}$ that are eventually zero is countable.
9. Show that $[0, \infty)$ and $\mathbb{R}$ are not homeomorphic.
10. Let $x = (x_1, x_2, \dots) \in \mathbb{R}^\omega$. Let $\{x^{(n)}\}_{n \in \mathbb{Z}_+}$ be a sequence of points in $\mathbb{R}^\omega$ such that for each coordinate $i \in \mathbb{Z}_+$, the sequence $\{x_i^{(n)}\}_{n \in \mathbb{Z}_+}$ converges to x_i in $\mathbb{R}$. Show that $\{x^{(n)}\}_{n \in \mathbb{Z}_+}$ converges to x if $\mathbb{R}^\omega$ has the product topology.
11. Give an example of a topological space that is connected but not path-connected.
12. Define what it means for a topological space X to be a Baire space.
13. Show that if a topological space X is Hausdorff, then a sequence in X can converge to at most one point in X .
14. Show that the function $f : [0, 2\pi) \rightarrow \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\}$ from the half-open interval to the circle defined by $f(t) = (\cos(t), \sin(t))$ is not a homeomorphism.
15. Let X be a locally compact Hausdorff space. Define the topology on $Y = X \cup \{\infty\}$ that makes Y the one-point compactification of X .